

Feedback Control Systems (EE3004)Date: June 3rd 2026

Course Instructor

Dr. Omer Saleem and Ms. Tamania Javaid

Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 05

Semester: SP-2026

Campus: Lahore

Dept: Electrical Engg.

Student Name

Roll No

Section

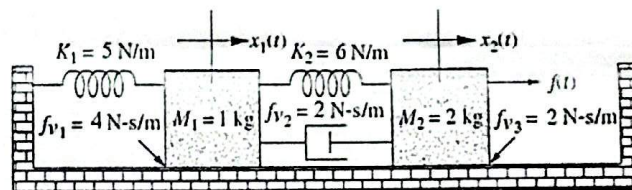
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CLO # 1: Develop mathematical models of physical systems using transfer functions and state-space methods.**Q1:****[20 marks]**

- a. Develop the state space representation of the following mechanical system. The variable $x_1(t)$ is the output.



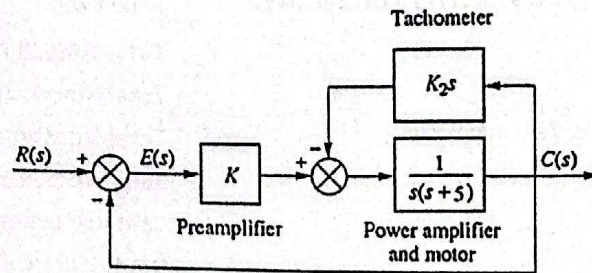
- b. Formulate transfer function of the form, $Y(s)/U(s)$, using the following state space representation. Hence, find the expression of output, $y(t)$, if the input signal, $u(t)$, is a unit-step signal.

$$\dot{x}(t) = \begin{bmatrix} 0 & 2 \\ -2 & -5 \end{bmatrix} x(t) + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 3 & 5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

CLO # 2: Analyze the transient and steady state response of a feedback system.**Q2:** The system shown below has its transient response altered by adding a tachometer.**[20 marks]**

- Reduce the block diagram to a unity feedback system and find the system's open-loop transfer function $G(s)$.
- Hence, reduce the unity feedback system in (a) to find the closed-loop transfer function $T(s)$.
- Design K and K_2 in the system to yield a damping ratio of 0.80 and a natural frequency of 12.0 rad/s.
- Use $G(s)$ in (a) to find the steady-state error of this system for a unit ramp input.
- What modification is needed in the pre-amplifier K to ensure the steady-state error becomes zero.



CLO # 3: Analyze the stability of a system using analytical methods.

Q3:

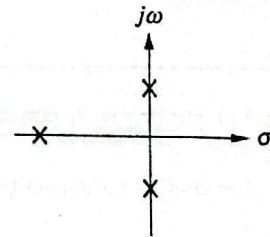
[20 marks]

- a. For the unity feedback system shown below, where:

$$G(s) = \frac{K}{(s+10)(s^2+4s+5)}$$

- Find the closed-loop transfer function, $T(s)$, of the system.
 - Hence, analyze $T(s)$ to calculate the range of K for stability.
- b. Find $T(s)$ and use Routh Hurwitz criteria to calculate K that will place the closed-loop poles of the system as shown in the Figure; one pole on real-axis and two poles on $j\omega$ -axis. The open loop transfer function $G(s)$ of unity feedback system is expressed below. Hence, find the value of the poles as well.

$$G(s) = \frac{Ks^2(s+1)}{s^3+s^2+2s+K}$$

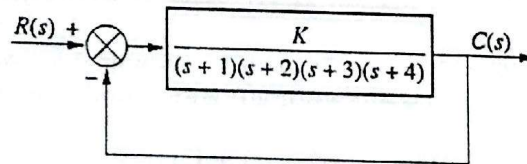


CLO # 4: Analyze a given system in frequency domain.

Q4:

[20 marks]

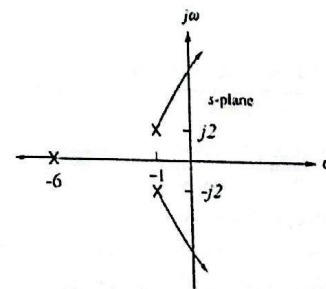
- a) For the system shown in Figure below,

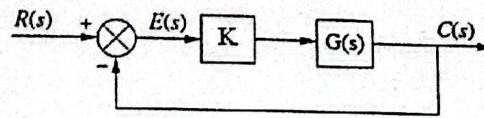


- Find the asymptotes (if any) of the root-locus.
- Find the break-away and break-in points (if any) of the root-locus.
- Find the crossings on the $j\omega$ -axis of the root-locus.
- Hence, illustrate the diagram of the root-locus.

- b) The root locus of a system, $G(s)$, is shown here:

- Find the open-loop transfer function, $G(s)$, of the system.
- Hence, using the appropriate property, find the gain K for which the system will yield a closed-loop pole on the real-axis at $s = -10$.





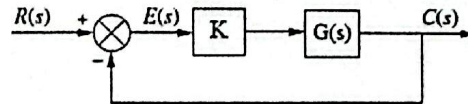
CLO # 5: Design a closed-loop feedback controller to meet transient and steady-state performance specifications.

Q5:

[20 marks]

The open loop transfer function of an un-compensated unity feedback system is given below.

$$K G(s) = \frac{K}{(s+1)(s+5)}$$



The dominant pole of the un-compensated system occurs at $K = 25$.

- Find the location of the dominant pole of the un-compensated system.
- Find the un-compensated system's settling-time, peak-time, and percentage overshoot.
- It is desired to compensate the system to decrease its settling-time by a factor of 2.0 without affecting its overshoot. Find the new settling-time, peak-time, and the new dominant pole location to satisfy the required performance metrics.
- Design a PD controller to satisfy the desired performance metrics. Write the transfer function of the controller including its pole/zeroes.
- Construct an op-amp circuit for the PD controller designed in (d). Use a capacitor of $10 \mu\text{F}$.